Power Budget

Team Number:	101
Project Name:	**Automatic Pet Feeder**
Team Member Names:	**Caleb Ross, Connor Erwin, Gary Hoffland, Daksh Raghav Preetha**

All Major Components	**Component Name**	**Part Number**	**olyVoltageRan**	**#**	**Absolute**	**talCurrent(mA)**	**Unit**
	Optical Sensor	TCST1103	1.2V to 70V	1	60	60	mA
	LED	151051RS11000	2.1V to 2.5V	2	30	60	mA
	Pushbutton	B3F-4055	0V to 24V	1	50	50	mA
	Opamp	MCP6004-I/P	1.8V to 6V	1	3	3	mA
	PIC Microcontroller Board	DM164150	1.8V to 5.5V	1	30	500	mA
	#NAME?	L7805CV	7.5V to 35V	1	1500	1500	mA
+5V Power Rail	**Component Name**	**Part Number**	**olyVoltageRan**	**#**	**MaximumCurre**	**talCurrent(mA)**	**Unit**
	Opamp	MCP6004-I/P	1.8V to 6V	1	3	3	mA
	Optical Sensor	TCST1103	1.2V to 70V	1	60	60	mA
	LED	151051RS11000	2V	2	30	60	mA
	Pushbutton	B3F-4055	0V to 24V	1	50	50	mA
	PIC Microcontroller Board	DM164150	1.8V to 5.5V	1	30	30	mA
					Subtotal	203	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	253.75	mA
c2. Regulator or Source Choice	+5V Regulator	LM7805CV	7.5V to 35V	1	1500	1500	mA
					Total Remaining Current Available on +5V Rail	1246.25	mA
External Power Source 1	**Component Name**	**Part Number**	**olyVoltageRan**	**Output**	**Absolute**	**Total**	**Unit**
Power Source 1 Selection	DC Power Jack	PJ-102AH	0V to 24V	1	5000	5000	mA
Power Rails Connected to External Power Source 1	5V Regulator	L7805CV	7.5V to 35V	1	253.75	253.75	mA
					Total Remaining Current Available on External Power Source 1	4746.25	mA
	Component Name	**Part Number**	**Supply**		**Capacity**	**Required**	
	DC Power Jack	PJ-102AH	0V to 24V		5000	253.75	mA
					Battery Life	19.7044335	hours